

Distributions

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Distributions

- ▶ A distribution is a generalized function.
- ▶ Distributions act as linear functionals on a set of test functions.
- ▶ A functional maps a set of functions to the reals.

Linear Functional

Distributions act as linear functionals on **test functions**.

Test functions are

- ▶ Smooth: infinitely many derivatives
- ▶ Compactly Supported: evaluate to zero outside a compact set

Linear Functional

Distributions act as **linear** functionals on test functions.
For distribution T , test functions f, g , and scalars a, b :

$$T[af + bg] = aT[f] + bT[g]$$

Linear Functional

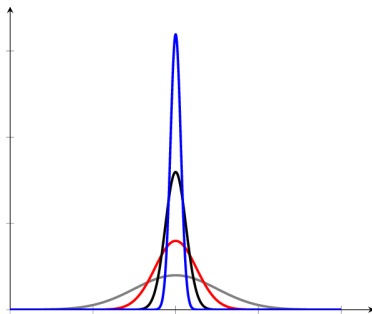
Distributions **act** as linear **functionals** on test functions.
To perform this action, we integrate the test function with the distribution function.

$$T[\phi] = \int_{-\infty}^{\infty} T(x)\phi(x)dx$$

The Dirac Delta "Function"

$$\int_{-\infty}^{\infty} f(x)\delta(x-a)dx = f(a)$$

Intuition: Gaussian, narrowing width.



Differential Equations

- ▶ Allow singular solutions to differential equations
- ▶ Used to build solutions to partial differential equations

Green's Functions

- ▶ L is a *Linear Differential Operator*
- ▶ G is a *Green's Function*
- ▶ f is a *Test Function*

$$\int LG(x)f(x)dx = \int \delta(x)f(x) = f(0).$$

Fourier Transforms

An example of a mapping, $f \rightarrow g$, where:

$$g(k) = \int_{-\infty}^{\infty} e^{i(2\pi k)x} f(x) dx$$

or, in the discrete case:

$$g(k) = \sum_{n=0}^{N-1} x_n e^{-i(2\pi k)n/N}$$

Dirac Delta Derivative

We will now find the weak derivative of the delta function.
Observe that:

$$\delta[f'] = \int_a^b \delta(x) f'(x) dx$$

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This means $\delta[f'] = -\delta'[f]$ so $\delta'[f] = -\delta[f']$.

Heaviside Function

We will now show that the weak derivative of the Heaviside function is the delta function.

The Heaviside function is defined as:

$$H(x) = \begin{cases} 0 & x < 0 \\ 1 & x > 0 \end{cases}$$

If the delta function is the weak derivative of the Heaviside function then:

$$\int_a^b H(x) f'(x) dx = H(x) f(x) \Big|_a^b - \int_a^b \delta(x) f(x) dx$$

Heaviside Function

We will show that the following holds under three cases:

$$\int_a^b H(x)f'(x)dx = H(x)f(x)\Big|_a^b - \int_a^b \delta(x)f(x)dx$$

These cases are:

1. $a < b < 0$
2. $a < 0 < b$
3. $0 < a < b$

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This indicates that $\frac{d}{dx}H[f(x)] = \delta[f]$ under certain definitions of derivation.

Questions?